BullSO 2026 Invitational
Astronomy C Answer Key

___/200

General Knowledge (2 each)

1. A star is observed with a parallax of 0.02 arcseconds and an apparent magnitude of +8.0. What is its absolute magnitude (M)?
   A. 3.0
   B. 4.5
   C. 5.0
   D. 6.5

2. In the "Ophion Star Family," many stars exhibit an "infrared excess" in their spectral energy distribution. What is the most likely physical cause?
   A. Relativistic Doppler shifting of the stellar spectra
   B. High concentrations of ionized Helium in the outer atmosphere
   C. Re-radiation of stellar light by a circumstellar dust disk
   D. The presence of a high-mass Black Hole companion

3. The "Janus" White Dwarf (ZTF J2033+32) shows a transition from Hydrogen-rich to Helium-rich atmospheric composition. Which process is hypothesized to cause this transition?
   A. Convective mixing of the thin H layer into the deeper He

layer
B. Accretion of Helium from a hidden M-dwarf companion
C. Photodissociation of Hydrogen into Helium via cosmic rays
D. Rapid rotation causing centrifugal separation of isotopes

4. Consider the equation for Hydrostatic Equilibrium. If the pressure gradient dP/dr becomes less negative while the density remains constant, what happens to the stellar structure?
   A. The star will contract to increase internal pressure
   B. The star will expand as gravity overcomes gas pressure
   C. The star will collapse into a singularity
   D. The star will achieve a new equilibrium at a lower temperature

5. Which of the following is a primary characteristic of WDJ181058.67+311940.94 that challenges standard cooling models?
   A. It is a "low-mass" white dwarf that should not have formed in the age of the universe
   B. It exhibits a magnetic field exceeding 500 Mega-Gauss
   C. It is a "metal-polluted" white dwarf with evidence of planetary debris
   D. It is an extremely massive white dwarf near the Chandrasekhar limit

6. In a non-relativistic, degenerate gas (like a White Dwarf core), what is the specific relationship between pressure (P) and density (rho)?

A. P is proportional to rho
B. P is proportional rho^4/3
C. P is proportional to rho^5/3
D. P is proportional to rho^2

7. A Mira variable's "period-luminosity" relationship is most accurately measured in which band to minimize the effects of interstellar extinction?
   A. Ultraviolet (U-band)
   B. Visual (V-band)
   C. K-band (Infrared)
   D. X-ray

8. For a star on the Asymptotic Giant Branch (AGB), what occurs during a "Thermal Pulse"?
   A. The core collapses into a neutron star
   B. The Helium shell ignites explosively in a thin layer
   C. The Hydrogen envelope is ejected to form a supernova
   D. The star transitions from the Hayashi track to the Henyey track

9. In the spectrum of a T Tauri star like HP Tau, the presence of the Lithium 670.8 nm absorption line indicates:
   A. The star has reached the Main Sequence
   B. The star is very young, as Lithium is easily destroyed in stellar interiors
   C. The star is extremely old and has undergone "dredge-up"
   D. The star is undergoing the CNO cycle

10. What is the Eddington Limit for a star?
    A. The maximum mass a star can have before electron degeneracy fails
    B. The luminosity at which radiation pressure outward equals gravity inward
    C. The minimum temperature required for Oxygen fusion
    D. The rotational velocity at which a star breaks apart

11. The Bone (G359.13) is a non-thermal filament near the Galactic Center. Its emission is dominated by synchrotron radiation, which implies:
    A. High-temperature thermal gas (10^7 K)
    B. Relativistic electrons spiraling in magnetic fields
    C. Reflection of light from Sagittarius A*
    D. Cold molecular gas (10 K)

12. Which of the following best describes the "S-process" of nucleosynthesis occurring in AGB stars?
    A. Rapid neutron capture in a supernova shockwave
    B. Slow neutron capture during the inter-pulse phase of Helium burning
    C. Proton capture in the core of a massive star
    D. Photodisintegration of Silicon into Iron

13. In the context of The Crab (M1), what is the "Pulsar Wind Nebula" (PWN)?
    A. The shell of gas ejected during the initial 1054 AD explosion
    B. A shock front where the pulsar's relativistic particles meet the ejecta

C. The accretion disk formed by a companion star
D. A region of cold dust blocking the pulsar's light

14. Type Ia Supernovae show a distinct "secondary maximum" in their light curves. In which wavelength range is this most prominent?
A. B-band (Blue)
B. V-band (Visual)
C. Near-Infrared (I or J band)
D. Gamma-ray

15. If the temperature of a Blackbody increases by a factor of 3, the total energy emitted per unit area (F) increases by a factor of:
A. 3
B. 9
C. 27
D. 81

16. Which of the following differentiates "Classical Cepheids" from "Type II Cepheids"?
A. Classical Cepheids are Population II stars found in the halo
B. Type II Cepheids are more luminous for the same period
C. Classical Cepheids are younger, Population I stars
D. Type II Cepheids do not follow a period-luminosity relation

17. What is the physical significance of the "Schönberg-Chandrasekhar Limit"?
A. It defines the maximum mass of a non-burning isothermal core

B. It defines the point where a star becomes a Black Hole
C. It is the maximum brightness of a Type Ia supernova
D. It is the limit of planetary nebula expansion

18. Why are O-type stars rarely found with planetary systems in the Orion Molecular Cloud?
A. They don't live long enough for planets to form
B. Their intense UV radiation evaporates protoplanetary disks
C. They only form in isolation, never in clouds
D. Their high gravity sucks all planetesimals into the star

19. In a "High-Mass X-Ray Binary" (HMXB), the donor star is typically:
A. An old, low-mass M-dwarf
B. A White Dwarf
C. An O or B-type Main Sequence star
D. A Red Giant on the AGB

20. The Helix Nebula (NGC 7293) exhibits "cometary knots." These are likely formed by:
A. Actual comets being vaporized by the central star
B. Rayleigh-Taylor instabilities between the hot wind and cool gas
C. Dark matter filaments
D. Magnetic reconnection events

21. What is the "triple-alpha" reaction's dependence on temperature (T)?
A. T^4

B. T^15
C. T^40
D. T^100

22. An RR Lyrae star has an apparent magnitude of 15.0. If its absolute magnitude is +0.5, what is the distance to the star?
A. 794 parsecs
B. 1,585 parsecs
C. 7,943 parsecs
D. 10,000 parsecs

23. Which of the following describes the spectrum of a Mira variable?
A. Strong emission lines of Hydrogen and TiO absorption bands
B. Purely continuum emission with no absorption
C. Broad Balmer lines indicating high pressure
D. Heavily redshifted Calcium lines

24. The Schwarzschild Radius of a 10 solar mass black hole is approximately:
A. 3 km
B. 30 km
C. 300 km
D. 3,000 km

25. HP Tau is a T Tauri star. Which of the following would be an expected feature of its light curve?
A. Extremely regular pulsations like a clock

B. Irregular flickers and "bursts" due to accretion
C. A single, sharp dip like an exoplanet transit
D. No variation whatsoever

26. In the Orion Molecular Cloud, "HII regions" are ionized primarily by:
A. X-rays from the central black hole
B. Lyman continuum photons from OB stars
C. Frictional heating from cloud collisions
D. Radioactive decay of Aluminum-26

27. What is the primary reason Type II supernovae (like Cas A) leave behind a compact object, while Type Ia do not?
A. Type Ia are more energetic and vaporize the entire star
B. Type Ia occur in lower density environments
C. Type II involve core collapse, whereas Type Ia involve thermonuclear runaway
D. Type II stars have more iron, which is magnetic

28. "Spectroscopic Parallax" is a distance-finding method that relies on:
A. Measuring the shift of a star against the background over 6 months
B. Comparing apparent magnitude to absolute magnitude derived from spectral type
C. Using the period of a pulsating star
D. Measuring the redshift of the star's spectral lines

29. The "Forbidden Lines" of [O III] (500.7 nm) seen in planetary nebulae can only occur because:
A. The gas is extremely dense
B. The gas is at a very high pressure
C. The gas is extremely tenuous, preventing collisional de-excitation
D. The central star is cooler than the Sun

30. Which of the following best describes "Coulomb Screening" in dense stellar cores?
A. The reduction of the repulsion between nuclei due to a "cloud" of electrons
B. The blocking of light by heavy metals
C. The process of making a star opaque to neutrinos
D. The way magnetic fields prevent fusion

31. Tycho's SNR is a "shell-type" remnant. In X-rays, the "forward shock" moves into the ISM. What does the "reverse shock" do?
A. It moves outward faster than the forward shock
B. It moves inward to heat the supernova ejecta
C. It causes the remnant to collapse into a black hole
D. It creates the pulsar at the center

32. The "CNO Cycle" is the dominant energy source in stars that are:
A. Cooler and less massive than the Sun
B. More massive than approximately 1.3 solar masses

C. In the pre-main sequence phase
D. Transitioning to the white dwarf phase

33. In the H-R diagram, where would you find a "Blue Straggler"?
A. Below the Main Sequence in the "sub-dwarf" region
B. On the Main Sequence, but above the turnoff point of its cluster
C. In the top right corner with the Red Supergiants
D. In the same region as the White Dwarfs

34. Why is CO (Carbon Monoxide) used as a tracer for H_2 in clouds like NGC 6559?
A. CO is more abundant than H_2
B. H_2 has no dipole moment and does not emit in the radio at low temperatures
C. CO is only found in star-forming regions
D. H_2 is blocked by the Earth's atmosphere

35. A pulsar's "Spin-down Age" is calculated based on:
A. The rate at which the pulsar's rotation period is increasing (P)
B. The temperature of the neutron star surface
C. The distance to the pulsar
D. The brightness of its optical pulses

36. Which of the following describes the "Horizontal Branch" phase?
A. Core Hydrogen burning

B. Core Helium burning
C. Core Carbon burning
D. Shell Helium burning only

37. The "Gamow Peak" represents:
A. The maximum temperature a star can reach
B. The energy at which nuclear fusion is most likely to occur
C. The highest point on an H-R diagram
D. The peak of the Planck curve

38. In the context of Cas A, what does "light echo" observation allow astronomers to do?
A. Predict when the next supernova will occur
B. See the spectrum of the original supernova explosion via reflected light
C. Measure the exact mass of the central neutron star
D. Calculate the speed of gravity

39. Which spectral type has the strongest Balmer (Hydrogen) absorption lines?
A. O
B. A
C. G
D. M

40. What defines a "T Tauri" variable star?
A. A star that has just finished core Helium burning
B. A pre-main sequence star with erratic variability and mass loss

C. A star that has reached the Chandrasekhar limit
D. A white dwarf that has cooled for over 10 billion years

Open Ended (3 each)

41. Explain the physical mechanism (the kappa-mechanism) that drives the periodic pulsations in Cepheid and RR Lyrae variables. Specifically, discuss how the ionization state of Helium in the star's atmosphere acts as a "valve" for radiation, and why this mechanism only functions effectively within a narrow range of effective temperatures on the H-R diagram.
kappa-mechanism: Explain that "kappa" refers to opacity. In the pulsation zone, Helium (He^+) is ionized to He^++ as the star contracts.
Energy Trap: Ionized Helium is more opaque, trapping radiation and increasing internal pressure, which pushes the outer layers outward.Expansion & Cooling: As the star expands, it cools, and the Helium recombines (He^++ back to He^+), becoming transparent and allowing radiation to escape.
Narrow Range: This only works in a specific temperature range where the Helium ionization zone is at the correct depth to drive the entire atmosphere.
42. Compare the visible light spectra of an O-type star and an M-type star. Identify the primary chemical markers (lines or bands) for each and explain why O-type stars lack the molecular bands found in M-type stars, even if they possess the same chemical abundance of those elements.

O-type Stars: High surface temperatures (>30,000 K) cause atoms to be highly ionized. Features include He II (ionized helium) lines.

M-type Stars: Cooler temperatures (<3,700 K) allow molecules to remain intact. Features include prominent Titanium Oxide (TiO) absorption bands.

Molecular Dissociation: Explain that O-type stars have the same elements, but the intense heat provides enough energy to break molecular bonds (dissociation), meaning molecules like TiO cannot exist.

43. Describe the structural changes inside a Sun-like star as it moves from the Main Sequence to the tip of the Red Giant Branch. Your answer should address the state of the core, the location of nuclear fusion, and the role of "electron

Core State: Core Hydrogen is exhausted, leaving an inert Helium core that begins to contract under gravity.

Shell Burning: Hydrogen fusion moves to a shell surrounding the core, which provides the high luminosity and pressure that expands the envelope.

Red Giant Growth: The envelope expands and cools (moving the star to the right), while the core becomes "electron degenerate."

Degeneracy: In the core, electrons are packed so tightly that the Pauli Exclusion Principle provides pressure independent of temperature, preventing further collapse until the Helium Flash.

44. Using a multi-wavelength approach, explain why the Crab Nebula (M1) looks fundamentally different in X-ray versus Radio observations. Discuss what these differences reveal about the energy of the electrons and the proximity of the emission to the central pulsar.

X-Ray (High Energy): Shows the "inner" nebula and the torus/jets. This emission is from the most energetic electrons recently accelerated by the pulsar.

Radio (Low Energy): Shows the "outer" and more diffuse nebula. These are older electrons that have migrated further away and lost energy over time.

Synchrotron Radiation: Both wavelengths are produced by electrons spiraling in magnetic fields; the difference in appearance is due to the electron's energy decay (synchrotron cooling).

45. Cold molecular clouds are the birthplaces of stars, yet they appear as dark voids in optical images. Explain the relationship between dust grain size and the "reddening" of light from stars behind or within the cloud. How do IR observations allow us to see "through" these clouds to identify protostars?

Mie Scattering/Extinction: Dust grains are roughly the size of blue light wavelengths, so they scatter/absorb blue light more effectively than red (reddening).

Optical Void: In the visible spectrum, the density of the dust is high enough to create infinite optical depth (total obscuration).

Infrared Advantage: IR wavelengths are much longer than the dust grain size, allowing them to pass through the cloud without being scattered.

Protostar Identification: IR allows us to see the thermal heat signature of the collapsing core (protostar) which is otherwise hidden.

46. If you are presented with two supernova spectra—one showing strong Balmer lines and the other showing no Hydrogen but strong Silicon absorption—identify the "Type" for each. Explain the evolutionary differences in the progenitor stars that lead to these two distinct spectral signatures.

Type II (Hydrogen Lines): Progenitor is a massive star (>8 M.) that retained its Hydrogen envelope before core collapse.

Type Ia (Silicon, No Hydrogen): Progenitor is a White Dwarf in a binary system that reached the Chandrasekhar limit. It has no Hydrogen because White Dwarfs are the exposed cores of former stars.

Evolutionary Path: Type II involves iron core collapse; Type Ia involves a thermonuclear runaway of Carbon/Oxygen.

47. T Tauri stars like HP Tau often exhibit an "infrared excess." Describe the physical structure surrounding a T Tauri star that produces this IR emission. How does this stage of evolution differ from the earlier "Protostar" phase in terms of mass accretion and hydrostatic equilibrium?

Physical Structure: The "infrared excess" is caused by a circumstellar/protoplanetary disk of gas and dust that absorbs starlight and re-emits it as heat (IR).

Evolutionary Stage: Unlike a protostar, a T Tauri star is visible in the optical, has cleared much of its "envelope," but is still contracting toward the Main Sequence.

Hydrostatic Equilibrium: T Tauri stars have achieved a degree of hydrostatic balance but are powered by gravitational contraction rather than Hydrogen fusion.

48. White dwarfs are often categorized by their atmospheric composition (e.g., DA vs. DB). Explain how "gravitational settling" influences the appearance of a white dwarf's spectrum over time. How do unusual objects like Janus challenge our understanding of how these atmospheres evolve as the star cools?

Gravitational Settling: Because gravity is so intense, heavy elements (metals) sink to the center, leaving the lightest elements (H or He) at the surface.

Spectral Types: DA white dwarfs have Hydrogen-rich atmospheres; DB have Helium-rich atmospheres.

Janus Transition: Janus challenges this by showing "patches" or transitions, suggesting that as a star cools, convective zones can reach deep enough to "dredge up" buried Helium and mix the atmosphere.

49. The Helix Nebula (NGC 7293) displays complex structures like "cometary knots." Discuss how the interaction between

the fast stellar wind from the hot central star and the slower, older material ejected during the AGB phase creates the distinct "bubbles" and filaments seen in planetary nebulae.

Interacting Winds: A fast, thin wind from the hot white dwarf core catches up to the slow, dense wind ejected earlier during the AGB phase (the "snowplow" effect).

Cometary Knots: These are dense clumps of molecular gas protected from the central star's ionizing radiation by their own shadows, creating "tails" pointing away from the star.

Ionization Fronts: The distinct "ring" shape is often an illusion of a 3D shell where the edges appear thicker from our line of sight.

50. In an X-ray binary system, matter is transferred from a donor star to a compact object (neutron star or black hole). Describe the process of "gravitational potential energy conversion" that occurs within the accretion disk. Why is the emission from this process primarily in the X-ray part of the spectrum rather than the visible?

Potential Energy Conversion: As matter falls from the donor star toward the compact object, it loses gravitational potential energy, which is converted into heat (kinetic energy).

Accretion Disk Friction: Friction between different layers of the disk (differential rotation) generates temperatures in the millions of Kelvin.

Wien's Law: Because the temperature is so high (10^6 - 10^7 K), the peak wavelength of emission shifts into the X-ray part of the electromagnetic spectrum.

Math (3 each)

51. A binary star system is observed to have an orbital period of 8 years and a semi-major axis of 4 AU. What is the total mass of the system in solar masses?

Answer: 1.0 solar masses. (Formula: Total Mass = a^3 / P^2 = 4^3 / 8^2 = 64 / 64 = 1).

52. A star in the Ophiuchus cloud complex has a measured parallax of 0.025 arcseconds. Calculate its distance in parsecs.

Answer: 40 pc. (Formula: d = 1 / parallax = 1 / 0.025).

53. A Type Ia supernova is observed in a distant galaxy with a peak apparent magnitude (m) of +15.7. Assuming the standard absolute magnitude (M) for a Type Ia supernova is -19.3, what is the distance modulus (m - M)?

Answer: 35.0. (Formula: 15.7 - (-19.3) = 35.0).

54. Based on the distance modulus of 35.0 from the previous question, calculate the distance to that galaxy in Megaparsecs (Mpc).

Answer: 100 Mpc. (Formula: d = 10^((35+5)/5) = 10^8 parsecs. 10^8 pc / 1,000,000 = 100 Mpc).

55. Star A and Star B have the same surface temperature. If Star A is 3 times more luminous than Star B, what is the ratio of Star A's radius to Star B's radius?

Answer: 1.73 to 1 (or square root of 3 to 1). (Since Luminosity is proportional to Radius squared, Radius is proportional to the square root of Luminosity).

56. A Cepheid variable star has a period of 10 days. Using the Period-Luminosity relation [M = -2.76 * log10(P) - 1.40], calculate its absolute magnitude.

Answer: -4.16. (Calculation: -2.76 * log10(10) - 1.40 = -2.76 - 1.40).

57. In a binary system, Star 1 is located 2 AU from the center of mass, and Star 2 is located 6 AU from the center of mass. What is the ratio of their masses (Mass 1 / Mass 2)?

Answer: 3. (Formula: M1 * r1 = M2 * r2, so M1/M2 = r2/r1 = 6/2).

58. If a star's surface temperature doubles while its radius is reduced to half its original size, by what factor does its total luminosity change?

Answer: It increases by a factor of 4. (Formula: L is proportional to R^2 * T^4. Calculation: (0.5)^2 * (2)^4 = 0.25 * 16 = 4).

59. An RR Lyrae star in a globular cluster has an apparent magnitude of 13.5. Given its absolute magnitude is +0.6, determine the distance to the cluster in parsecs.

Answer: 3,802 parsecs. (Formula: d = 10^((m - M + 5) / 5) = 10^((13.5 - 0.6 + 5) / 5) = 10^3.58).

60. A star's spectrum peaks at a wavelength of 290 nm. According to Wien's Law (Constant = 2,900,000 nm*K), what is the star's surface temperature in Kelvin?

Answer: 10,000 K. (Formula: T = 2,900,000 / peak wavelength = 2,900,000 / 290).

61. Two stars in a circular orbit have a separation of 10 AU and an orbital velocity of 20 km/s relative to each other. Calculate the orbital period in years.

Answer: Approximately 15 years. (Circumference = 2 * pi * 10 AU = 9.42 billion km. Time = Distance / Velocity = 471 million seconds. Divide by 31.5 million seconds per year).

62. An eclipsing binary system shows maximum radial velocity shifts of Velocity A = 30 km/s and Velocity B = 10 km/s. What is the mass ratio (Mass A / Mass B)?

Answer: 0.33. (Formula: Ma / Mb = Vb / Va = 10 / 30).

63. A G2V star (Sun-like, Absolute Magnitude = +4.8) is observed with an apparent magnitude of +9.8. What is its distance in parsecs?

Answer: 100 pc. (Formula: d = 10^((9.8 - 4.8 + 5) / 5) = 10^2).

64. If a star has a luminosity of 100 solar luminosities and a surface temperature twice that of the Sun (2 * T-sun), calculate its radius in terms of solar radii.

Answer: 2.5 solar radii. (Formula: 100 = R^2 * 2^4. 100 = 16 * R^2. R^2 = 6.25. R = 2.5).

65. Calculate the density of a white dwarf with a mass of 1.0 solar mass (2 * 10^30 kg) and a radius of 6,371 km (6.37 * 10^6 meters). Give your answer in kg/m^3.

Answer: 1.84 * 10^9 kg/m^3. (Volume = 4/3 * pi * r^3 = 1.08 * 10^21 m^3. Density = Mass / Volume).

66. A star is receding from Earth at 300 km/s. If the H-alpha rest wavelength is 656.3 nm, what is the observed Doppler-shifted wavelength? (Speed of light = 300,000 km/s).

Answer: 656.96 nm. (Shift = wavelength * (v / c) = 656.3 * 0.001 = 0.6563. New wavelength = 656.3 + 0.6563).

67. Three stars of equal mass (M) move in a circular orbit of radius (R) about their common center of mass, maintaining an equilateral triangle. What is the orbital velocity of each star?

Answer: v = square root of [G * M / (R * square root of 3)]. (Balanced gravitational pull of two stars against centripetal force).

68. According to Kepler's Third Law, if a planet's orbital distance from its star is increased by a factor of 4, by what factor does its orbital period increase?

Answer: 8. (Formula: P^2 is proportional to a^3. So P is proportional to a^1.5. 4^1.5 = 8).

69. A star has a visual absolute magnitude of +1.0 and a bolometric correction (BC) of -2.0. Calculate its bolometric absolute magnitude (M-bol).

Answer: -1.0. (Formula: M-bol = Visual Magnitude + BC = 1.0 - 2.0).

70. A pulsar has a period (P) of 0.033 seconds and a spin-down rate (P-dot) of 4.2 * 10^-13 s/s. Calculate the characteristic age (P / 2 * P-dot) in years.

Answer: Approximately 1,245 years. (Calculation: 0.033 / (8.4 * 10^-13) = 39.2 billion seconds. 39.2 billion / 31.5 million seconds per year = 1,245).

DSO (2 each)

71. Identify the object in Image A.

Answer: Helix Nebula (NGC 7293)

72. Based on the spectral features of the central star in Image A, what is the primary source of the ionizing radiation that causes the nebula to glow?

Answer: High-energy Ultraviolet (UV) photons from the extremely hot, exposed core (pre-white dwarf).

73. On an H-R diagram, where would the central star of the object in Image A be located compared to its previous Red Giant phase?

Answer: It has moved horizontally to the left, toward much higher temperatures at a constant or slightly decreasing luminosity.

74. Identify the object in Image B.

Answer: Tycho's SNR

75. Light curves of the original event that created the object in Image B showed a lack of Hydrogen but a strong Silicon II absorption line at 615 nm. Classify this supernova.

Answer: Type Ia Supernova.

76. Describe the "thermonuclear runaway" process that occurred in the white dwarf progenitor of the object in Image B.

Answer: Carbon fusion ignited under degenerate conditions; the temperature skyrocketed without a correlating expansion, leading to a massive explosion that destroyed the star.

77. Identify the object based on the Spectrum in Graph C.

Answer: Janus (ZTF J2033+32)

78. If you were to observe the two different sides of the object in Graph C, which specific physical process is believed to prevent the Hydrogen and Helium from mixing?

Answer: Strong, asymmetric magnetic fields that inhibit convection on one hemisphere.

79. How does the rotation of the object in Graph C affect its observed light curve?

Answer: It produces a periodic variation in brightness and spectral line strength as the two different chemical hemispheres rotate into view.

80. Identify the object in Image D.

Answer: Sharpless 29 (NGC 6559)

81. What is the specific chemical transition responsible for the red glow seen in the emission portion of the object in Image D?

Answer: The H-alpha transition (n=3 to n=2) in neutral Hydrogen atoms.

82. Explain why "cold molecular clouds" within the region shown in Image D appear as dark silhouettes against the bright background.

Answer: The clouds have high optical depth due to dust, which absorbs and scatters all visible background light.

83. Identify the object based on the Light Curve in Graph E.

Answer: Mira (Omicron Ceti)

84. The object in Graph E exhibits "shock-driven" emission lines. What physical process in the star's evolution creates these shocks?

Answer: Large-scale atmospheric pulsations where the star physically expands and contracts, sending shockwaves through its bloated envelope.

85. Mira variables like this one are located on which branch of the H-R diagram?

Answer: The Asymptotic Giant Branch (AGB).

86. Identify the object in Image F.

Answer: The Crab Nebula (M1)

87. Using multi-wavelength images, the object in Image F appears "filled" in X-rays but "filamentary" in optical. What is the source of the filled X-ray emission?

Answer: A Pulsar Wind Nebula (PWN) consisting of relativistic particles accelerated by the central pulsar's magnetic field.

88. What is the primary method of energy loss for the central pulsar in Image F that causes its rotation period (P) to increase over time?

Answer: Magnetic dipole radiation and the acceleration of the pulsar wind.

89. Identify the object based on Image G.

Answer: HP Tau

90. What is the primary source of luminosity for this star?

Answer: Gravitational contraction (Kelvin-Helmholtz mechanism), as it has not yet reached the temperatures required for core Hydrogen fusion.

91. Stars like the one in Image G are known for "mass loss." How is this mass loss typically observed in their spectra?

Answer: Through P-Cygni profiles, which show blue-shifted absorption and red-shifted emission lines.

92. Identify the object in Image H.

Answer: The Bone (G359.13)

93. If the object in Image H emits via synchrotron radiation, what must be true about the velocity of the electrons within it?

Answer: They must be moving at relativistic speeds (close to the speed of light).

94. Why is the object in Image H invisible in optical images but prominent in Radio and X-ray?

Answer: Massive amounts of interstellar dust block visible light from that region, while longer Radio waves and high-energy X-rays can pass through.

95. Identify the object in Image I.

Answer: Cas A (Cassiopeia A)

96. Spectroscopic analysis of the "knots" in the object in Image I shows they are rich in Oxygen, Neon, and Silicon. What does this tell us about the progenitor star?

Answer: The progenitor was a massive star that had completed advanced stages of nuclear burning (nucleosynthesis) before exploding.

97. What process in the expanding shockwave of the object in Image I produces its prominent radio emission?

Answer: The shockwave accelerates electrons to relativistic speeds, which then interact with magnetic fields (synchrotron emission).

98. Identify the object based on the Spectrum in Graph J.

Answer: WDJ181058.67+311940.94

(Note: Graph J should show spectral lines split into multiple components).

99. How do the strong magnetic fields of the object in Graph J affect the observed spectral lines?

Answer: They cause the spectral lines to split into multiple components via the Zeeman Effect.

100. Because the object in Graph J is "metal-polluted," what does its Blackbody radiation curve reveal about its environment?

Answer: It likely has an infrared excess, indicating a disk of debris (likely the remains of destroyed planets) orbiting the star.

BullSO, University of South Florida - Repaired practice edition - Answer key

Division C | Astronomy | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (6 points) Authored written substitute for unavailable source material: Solve this case and explain your reasoning: A source has m = 10 and M = 5, with negligible extinction. Find distance and compare its flux with the same source at 10 pc.

reasoning: Distance modulus is 5.
d = 10 x 10^(5/5) = 100 pc.
At ten times the distance, flux is one hundredth of its 10 pc value. (3 points).

conclusion: The distance is 100 pc, consistent with a five-magnitude dimming. (3 points).